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Conference on Algebraic Representation Theory 2026

CART 2026

December 4 (Fri.) – 6 (Sun.), 2026

Okinawa Institute of Science and Technology (OIST)
Seminar Room L4E48, Okinawa, Japan

Program

	December 4 (Fri.)	December 5 (Sat.)		December 6 (Sun.)
9:30 – 10:30	You-Hung Hsu	Haruto Murata	9:30 – 10:30	Yumeng Wu
10:30 – 11:00	(tea break)	(tea break)	10:30 – 10:45	(tea break)
11:00 – 12:00	Katsuyuki Bando	Chun-Ju Lai	10:45 – 11:45	Jiepeng Fang
12:00 – 14:00	Lunch	Lunch	11:45 – 13:00	Lunch
14:00 – 15:00	Donggun Lee	Kang Lu	13:00 – 14:00	Pavel Turek
15:00 – 15:20	(tea break)	(tea break)		
15:20 – 16:20	Woo-Seok Jung	Yung-Ning Peng		
16:20 – 16:40	(tea break)	(tea break)		
16:40 – 17:40	Heizo Sakamoto	Dong Jun Choi		
18:00 –		Banquet		

* The Zoom link will be shared later.

Titles and Abstracts

December 4 (Fri.)

You-Hung Hsu (National Cheng Kung University)

Title: 0-Affine Quantum Groups, K-Theoretic Hall Algebras, and Categorical Actions

Abstract: In this talk, I will discuss the 0-affine quantum group and its relation to K-theoretic Hall algebras. In particular, the positive part of the 0-affine quantum group can be realized as the K-theoretic Hall algebra of a type A Dynkin quiver. I will then describe a categorical action of this positive part and show how such an action induces semiorthogonal decompositions on the corresponding weight categories. The main geometric examples are the bounded derived categories of coherent sheaves on partial flag varieties. If time permits, I will also mention some related questions and further directions.

Katsuyuki Bando (Keio University)

Title: Derived geometric Satake equivalence in mixed characteristic

Abstract: The geometric Satake equivalence is the categorification of the Satake isomorphism. Its derived version, "the derived geometric Satake equivalence, " is proved by Bezrukavnikov–Finkelberg in the case of equal characteristic. In this talk, we will explain a new affine Grassmannian which connects an equal characteristic affine Grassmannian and Zhu's Witt-vector affine Grassmannian. As a result, we deduce the derived Satake equivalence in mixed characteristic. By the same argument, we also obtain the mixed characteristic version of Bezrukavnikov's equivalence between two categorifications of an affine Hecke algebra. Moreover, we explain the derived geometric Satake equivalence on the Beilinson–Drinfeld Grassmannian with one leg in mixed characteristic.

Donggun Lee (Korea Institute for Advanced Study)

Title: Hessenberg character

Abstract: Hessenberg varieties are subvarieties of flag varieties whose cohomology comes with a natural Weyl group action. In type A, these representations are closely connected to positivity questions in the Stanley-Stembridge and Haiman conjectures. This raises the question of whether such positivity phenomena extend naturally to other Lie types. A first step is to compute these Weyl group representations.

After briefly reviewing Hessenberg varieties and our recent results relating them to Lusztig varieties, I will present our construction of the Hessenberg functor on the Hecke category. This functor gives rise to a representation-valued trace on the Hecke algebra, called the Hessenberg character, which recovers the graded cohomology representations of a natural class of Hessenberg varieties. We then identify the Hessenberg functor with a microlocalization of the functor underlying Lusztig's character, allowing us to use his theory of character sheaves and nonabelian Fourier transform to obtain an explicit character formula.

Based on joint work with Patrick Brosnan and Jaehyun Hong.

Woo-Seok Jung (University of Seoul)

Title: A Comparison of Cluster Algebra Structures Arising from i-Boxes and Demazure Weaves

Abstract: We compare two cluster algebra structures associated with a positive braid of finite ADE type. The first arises from a localized bosonic extension, whose initial seeds are constructed from admissible chains of i-boxes and are closely related to monoidal categorification. The second arises from the coordinate ring of a braid variety, with initial seeds constructed from Demazure weaves.

For each admissible chain of i-boxes, we explicitly construct a corresponding Demazure weave and show that the resulting seeds are identified under an algebra isomorphism between the localized bosonic extension and the coordinate ring of the braid variety. Under this isomorphism, the PBW vectors are identified with the natural coordinates on the braid variety.

This is joint work with JiSun Huh, Myungho Kim, and Euiyong Park.

Heizo Sakamoto (The University of Tokyo)

Title: Classification of real and imaginary modules of quantum affine algebras in monoidal categorifications of affine cluster algebras

Abstract: A finite-dimensional module L over a quantum affine algebra is called real if the tensor product $L \otimes L$ is irreducible. Classifying real modules is a fundamental problem in understanding the tensor product structure of the category of finite-dimensional modules. For suitable subcategories $\mathcal{C}$, the Grothendieck ring $K(\mathcal{C})$ admits a cluster algebra structure in which cluster monomials correspond to irreducible modules. Hernandez and Leclerc conjectured that, in such a category, an irreducible module corresponds to a cluster monomial if and only if it is real. This conjecture is known in finite type but remains open in general. In this talk, I will construct subcategories that provide monoidal categorifications of cluster algebras of affine type and show that the conjecture holds in these categories. I will also briefly discuss the classification of prime modules in this setting.

December 5 (Sat.)

Haruto Murata (The University of Tokyo)

Title: Quantum imaginary Schur–Weyl duality

Abstract: Khovanov–Lauda–Rouquier algebras of affine type A depend on a parameter t and, at $t = 1$, are isomorphic to affine Hecke algebras of type A. The representation theory of Khovanov–Lauda–Rouquier algebras is reduced to the study of quotient algebras defined for each positive root of affine type A via stratification, among which those for imaginary roots are substantially more complicated than those for real roots. When $t = 1$, the imaginary strata are related to Schur algebras through the imaginary Schur–Weyl duality established by Kleshchev and Muth. In this talk, we generalize this duality to an arbitrary parameter t and establish a connection with t-Schur algebras. As an application, we explain how irreducible characters are related to the canonical basis.

Chun-Ju Lai (Academia Sinica)

Title: Foundations and Applications of Quantum Wreath Products

Abstract: Quantum wreath products deform classical wreath product groups by allowing relation coefficients to take values in a tensor product algebra rather than the ground field. This framework unifies the study of structure and representation theory of algebras admitting a Bernstein–Lusztig presentation, including Vignéras' pro-p Hecke algebras for p-adic groups (and their mteaplectic analogs), Kleshchev–Muth's affine zigzag algebras, and Hu's "Hecke algebras" for the wreath products between symmetric groups. We will discuss recent progress of our theory in seemingly unrelated directions: the Ginzburg–Guay–Opdam–Rouquier conjecture on highest-weight covers, the realization of imaginary strata for affine KLR algebras in positive characteristic, the breakdown of the local Shimura correspondence at the pro-p level, and the categorification perspectives of towers of quantum wreath products. The talk is based on joint work with Buciumas (Pohang), Cailan Li (Academia Sinica), Minets (Bonn), Nakano (Georgia), and Ziqing Xiang (Shenzhen).

Kang Lu (Southern University of Science and Technology)

Title: Drinfeld presentations of twisted Yangians and applications

Abstract: Twisted Yangians are coideal analogues of Yangians associated with symmetric pairs, with connections to representation theory and integrable systems. In this talk, I will discuss recent developments in their Drinfeld current presentations for all split and quasi-split types. Starting from twisted current algebras, I will explain the construction of twisted Yangians and their minimalistic presentations. I will then describe explicit embeddings into the Yangians, which establish the right coideal structures and connect the Drinfeld and J-presentations. These structural results provide tools for studying representations and applications of twisted Yangians. I will also discuss selected recent applications.

Yung-Ning Peng (National Cheng Kung University)

Title: Shifted twisted Yangians and finite W-algebras

Abstract: It is well known that the finite W-algebra associated to an arbitrary nilpotent element in a general linear Lie algebra is isomorphic to the corresponding truncated shifted Yangian due to Brundan–Kleshchev. Such an isomorphism is known to exist for BCD types of rectangular nilpotent elements, in which the role of the Yangians should be replaced by the corresponding *twisted* Yangians.

In this talk, we will introduce the notion of parabolic presentations for twisted Yangians, which interpolate between the R-matrix and Drinfeld presentations. This allows one to define the *shifted* twisted Yangians and their quotient algebras known as truncated shifted twisted Yangians. We further prove that our truncated shifted twisted Yangian is isomorphic to the finite W-algebra associated with every even nilpotent element in type B and C, as well as some special cases of type D. We conjecture the existence of a certain Pfaffian-like element in finite W-algebras of type D, whose existence will imply that the isomorphism mentioned above exists for type D as well. This talk is based on joint work with Kang Lu, Lukas Tappeiner, Lewis Topley, and Weiqiang Wang.

Dong Jun Choi (Seoul National University)

Title: Poisson Structures of Deformed and Classical W-algebras

Abstract: Deformed W-algebras have close connections with quantum affine algebras and q-characters. In type A, their Poisson versions admit a description in terms of q-difference operators, paralleling the description of classical W-algebras by differential operators.

In this talk, we present an algebraic formulation of q-difference Drinfeld–Sokolov reduction and explain how it produces the deformed W-algebra of type A. Using determinant formulas and the Miura transformation, we construct explicit linear combinations of the deformed W-algebra generators whose formal limits recover the standard generators of the classical W-algebra. We also show that the Poisson brackets recover the classical Poisson structure under this limit.

This talk is based on joint work with Uhi Rinn Suh.

December 6 (Sun.)

Yumeng Wu (Peking University)

Title: PBW and Canonical Bases for Tame Quivers: Geometry of Transition Coefficients

Abstract: This talk concerns the geometric meaning of the transition coefficients between PBW bases and the canonical basis of quantum groups associated with tame quivers, with a focus on the Kronecker quiver. After recalling Lusztig's geometric construction, I will explain a realization of PBW basis elements by sheaf complexes on representation varieties. The coefficients expressing canonical basis elements in the PBW basis can then be described through graded multiplicities of local systems in restrictions of intersection cohomology complexes to suitable strata. I will discuss how purity enters this description and how the resulting geometric interpretation accounts for the triangularity of the transition matrix and the positivity of its coefficients. The talk is based on joint work with Jie Xiao.

Jiepeng Fang (The University of Hong Kong)

Title: Thickening realization and positivity properties of canonical bases

Abstract: For any symmetric Cartan datum, Lusztig conjectured that the canonical basis $\dot{\mathbf{B}}$ of the modified quantum group $\dot{\mathbf{U}}$ has positivity properties with respect to the multiplication and comultiplication. We introduce an algebraic method called the thickening realization to relate $\dot{\mathbf{B}}$ to the canonical basis of the negative part $\tilde{\mathbf{U}}^-$ of a larger quantum group. More precisely, it identifies the structure constants in $\dot{\mathbf{U}}$ with relevant structure constants in $\tilde{\mathbf{U}}^-$. We prove that $\dot{\mathbf{B}}$ has the positivity properties with respect to the comultiplication and bilinear form of $\dot{\mathbf{U}}$, as well as the multiplication whenever one factor is spherical parabolic. In particular, Lusztig's positivity conjecture holds for any simply-laced finite type. This is a joint work with Xuhua He.

Pavel Turek (Okinawa Institute of Science and Technology)

Title: New decomposition columns in every block

Abstract: A central open problem in the modular representation theory of symmetric groups (and Hecke algebras) is the computation of decomposition numbers, which describe how characteristic-zero irreducible modules decompose in prime characteristic. These numbers are organised in a block matrix, called the decomposition matrix, whose structure remains a mystery. In this talk, we present new whole columns of this matrix labelled by a novel family of partitions called d-balanced partitions; this seems to be the first description of whole columns in each block to date! Moreover, the column description is exceptionally elementary. The talk is based on results proved with David Hemmer (for $d = 2$) and with Bim Gustavsson, Stacey Law, Lorenzo Putignano and Liron Speyer (for $d > 2$).